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Products of three triangular matrices

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Abstract

We prove that any square matrix over a field is the product of at most three triangular matrices. We also give explicit $\mathcal{LU}$ factorizations for all 2×2 and 3×3 matrices over fields. © 1999 Published by Elsevier Science Inc. All rights reserved.

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0. Introduction

Triangular matrices have been considered from the times of Frobenius (see Frobenius, Collected Papers VI. 4, p. 548). If K is any field with $|K| \geq 3$, the group of nonsingular lower triangular $n \times n$ matrices is an easily defined solvable group (see [11, p. 16]), and those with all diagonal entries equal to 1, a nilpotent group. The theorems of Lie–Kolchin [11, 5.8] and Kolchin–Malcev [5, 11, 3.6] show the importance attached to triangular matrices.

Linear equations have particularly transparent solutions when the matrix of coefficients is either lower triangular or upper triangular. Forsythe [2], Gantmacher [3], Choleski and several authors have given criteria for $\mathcal{LU}$ and $\mathcal{LU}$ decompositions for a matrix. These are found useful in Numerical Analysis.

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It is a fairly easy problem to show that any $n \times n$ matrix is a finite product of triangular matrices (at most $3n - 2$) (see [7, Problem 928]).

Fong and Sourour have shown that any matrix A with $\det A = 1$ is a product of at most three unipotent matrices (see [1]).

We prove in this paper that every square matrix over a field is the product of at most three triangular matrices Theorem 1.3. There is a 3×3 matrix A such that $A \neq LUM$, where L, M are the lower triangular matrices with diagonal entries equal to 1 and U is an upper triangular matrix Remark 3.3. Vaserstein and Wheland [10] proved in 1990 that every invertible matrix over a ring of stable rank 1 can be written as $\lambda\rho\mu$, where λ, μ are the lower triangular matrices and ρ an upper triangular matrix. In [10, p. 267] they remarked that the diagonal entries of λ and μ can be taken to be 1. Hence their remark does not hold for singular matrices and hence their method of proving $\mathcal{LU}$ decomposition is not likely to carry over to singular matrices.

In Section 2, we give explicit $\mathcal{LU}$ decompositions for all 2×2 and 3×3 matrices over a field and this can be used as a ready reckoner. In engineering problems quite often Crout's method is used to solve linear equations in three variables obtaining $\mathcal{LU}$ decomposition of the coefficient matrix. But this method cannot always work since there are matrices not allowing $\mathcal{LU}$ decomposition.

Throughout this paper F is a field, $M_n(F)$ is the set of all $n \times n$ matrices with coefficients in F . If $A \in M_n(F)$ then we write $A = (a_{ij})$, $a_{ij} \in F$, $1 \leq i, j \leq n$. $\mathcal{L}$ denotes the set of all lower triangular matrices, i.e., $\mathcal{L} = \{A = (a_{ij}) : a_{ij} = 0 \text{ whenever } i < j\}$. $\mathcal{U}$ denotes the set of all upper triangular matrices, i.e.,

$$\mathcal{U} = \{A : A = (a_{ij}) : a_{ij} = 0 \text{ whenever } i > j\}.$$

1. $\mathcal{LU}$ decomposition

Theorem 1.1 [10]. *Let $A \in M_n(F)$ be nonsingular. Then $A = LUM$, where L, M are the lower triangular matrices in $M_n(F)$ and U is an upper triangular matrix in $M_n(F)$.*

The following proposition is essential for our proof. It is a standard result. For the sake of completeness we provide the proof.

Proposition 1.2. *Let $A \in M_n(F)$. Then there exists a nonsingular $P \in M_n(F)$ and $L \in \mathcal{L}$ such that $A = PL$.*

Proof. We shall show that there exists a nonsingular $n \times n$ matrix P such that PA is lower triangular. We shall use induction on n , the case $n = 1$ being trivial. Let the result be true for $(n - 1) \times (n - 1)$ matrices. Let $A \in M_n(F)$.

Case (i): Suppose $A = (a_{ij})$ is such that $a_{in} = 0$ for $1 \leq i \leq n-1$.

Apply induction assumption on the $(n-1) \times (n-1)$ matrix $A' = (a_{ij})$, $1 \leq i, j \leq n-1$. Accordingly there is a nonsingular $(n-1) \times (n-1)$ matrix $Q = (q_{ij})$ such that QA' is lower triangular. Let $P = (p_{ij})$, $1 \leq i \leq n$, $1 \leq j \leq n$ be the $n \times n$ matrix where $p_{nn} = 1$, $p_{in} = 0 = p_{ni}$ for $1 \leq i \leq n-1$ and $p_{ij} = q_{ij}$ in all other cases. Then P has determinant equal to determinant of Q and PA is lower triangular.

Case (ii): Suppose for some $k \leq n-1$, we have $a_{kn} \neq 0$. Let us denote by $e_1, \dots, e_n$ the 1st, 2nd, ..., n th columns of the identity matrix. Define $Q = (q_{ij})$, $1 \leq i, j \leq n$ as follows: if $j \neq k, n$ then the j th column of Q is e_j and the n th column of Q is e_k . For the k th column of Q , we set $q_{kk} = -a_{nn}/a_{kn}$, $q_{nk} = 1$ and if $i \neq k, n$, $q_{ik} = -a_{in}/a_{kn}$. It is easy to verify that the determinant of $Q = -1$ so that Q is nonsingular. Further if $QA = B = (b_{ij})$, one verifies that $b_{in} = 0$ for $1 \leq i \leq n-1$. By Case (i), we can find a nonsingular matrix P' such that $P'B = P'QA$ is a lower triangular matrix L . We merely take $P = (P'Q)$. Now the assertion follows by induction and then the proposition easily follows. $\square$

Theorem 1.3. Let $A \in M_n(F)$, $n \geq 2$ be an arbitrary matrix. Then there exist $L, M \in \mathcal{L}$, $U \in \mathcal{U}$ such that L, U are nonsingular and such that $A = LUM$. Moreover we can arrange that in L, U , all the diagonal entries are 1.

Proof. If A is nonsingular then Theorem 1.1 completes the proof. Suppose A is a singular matrix. By Proposition 1.2 there exists a nonsingular $P \in M_n(F)$ and $L_1 \in \mathcal{L}$ such that $A = PL_1$. By Theorem 1.1 $P = LUM$, where $L, M \in \mathcal{L}$ and $U \in \mathcal{U}$. Now $A = LU(ML_1)$ and $ML_1 \in \mathcal{L}$. Hence the theorem follows.

When A is nonsingular and $A = LUM$ then L, U and M are all nonsingular. When A is singular and $P = LUM$, L, U are nonsingular and hence in both cases we get $A = LUM$ with L, U nonsingular.

Suppose $A = LUM$, with L, U nonsingular. Let $L = (b_{ij})$. Now $b_{ii} \neq 0$ for each $i = 1, 2, \dots, n$. Let $D_1 = \text{diag}(b_{11}^{-1}, b_{22}^{-1}, \dots, b_{nn}^{-1})$ the diagonal matrix. Then $D_1^{-1} = \text{diag}(b_{11}, b_{22}, \dots, b_{nn})$. Now $A = (LD_1)(D_1^{-1}U)M$. $LD_1 \in \mathcal{L}$ and all its diagonal entries are 1. Also $D_1^{-1}U \in \mathcal{U}$. Let $D_1^{-1}U = (c_{ij})$. $D_1^{-1}U$ is also nonsingular so that $c_{ii} \neq 0$ for each $i = 1, 2, \dots, n$. Let $D_2 = \text{diag}(c_{11}^{-1}, c_{22}^{-1}, \dots, c_{nn}^{-1})$. Now $D_1^{-1}UD_2 \in \mathcal{U}$ and all its diagonal entries are 1. $D_2^{-1} = \text{diag}(c_{11}, c_{22}, \dots, c_{nn}) \in \mathcal{L}$. Hence $D_2^{-1}M \in \mathcal{L}$. Thus $A = (LD_1)(D_1^{-1}UD_2)(D_2^{-1}M)$ is the required arrangement. This completes the proof of the theorem. $\square$

Remark 1.4. One of the referees has observed that instead of $\mathcal{L}\mathcal{U}\mathcal{L}$ decomposition we can equivalently consider $\mathcal{U}\mathcal{L}\mathcal{U}$ decomposition. Let us then take the proposition granted for nonsingular A (e.g. by Vaserstein and Wheland taking transpose). It is a standard result that for any matrix A there is a permutation matrix P , a lower triangular matrix L with unit diagonal and an

upper triangular matrix U such that $PA = LU$, see e.g [8, p. 57]. Here both P and L are nonsingular, so $A = QU$ where Q is nonsingular. Now applying $\mathcal{ULU}$ decomposition for Q we get a $\mathcal{ULU}$ decomposition for A .

2. Decompositions for 2×2 and 3×3 matrices

Proposition 2.1. *Let*

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

be in $M_2(F)$ with $b \neq 0$.

(i) *If $a \neq 0$ then*

$$A = \begin{pmatrix} 1 & 0 \\ \frac{c}{a} & d - \frac{bc}{a} \end{pmatrix} \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}.$$

(ii) *If $d \neq 0$, then*

$$A = \begin{pmatrix} a - \frac{bc}{d} & \frac{b}{d} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ c & d \end{pmatrix}.$$

(iii)

$$\begin{pmatrix} 0 & b \\ c & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -c & 0 \\ c & b \end{pmatrix}.$$

Proof. Is a straight forward verification. $\square$

We now consider 3×3 matrices. The decompositions we explicitly give can be checked to be correct by calculating the product.

Let $A \in M_3(F)$, $a = (a_{ij})$ with $a_{ij} \in F$, $1 \leq i, j \leq 3$.

Proposition 2.2. *Suppose $a_{13} \neq 0$. Put*

$$\alpha = \frac{a_{11}a_{23} - a_{13}a_{21}}{a_{13}}, \quad \beta = \frac{a_{12}a_{23} - a_{13}a_{22}}{a_{13}},$$

$$\gamma = \frac{a_{11}a_{33} - a_{13}a_{31}}{a_{13}}, \quad \delta = \frac{a_{12}a_{33} - a_{13}a_{32}}{a_{13}}.$$

Then $\alpha\delta - \beta\gamma = \det A$ holds.

(i) *If $\beta = 0$, then*

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \frac{a_{23}}{a_{13}} - 1 & 1 & 0 \\ \frac{a_{33}}{a_{13}} & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha & 0 & 0 \\ \gamma & \delta & 0 \\ -\alpha - \gamma + a_{11} & -\delta + a_{12} & a_{13} \end{pmatrix}.$$

(ii) If $\beta \neq 0$ and $\det A = 0$, then

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \frac{a_{23}}{a_{13}} & 1 & 0 \\ \frac{a_{33}-1}{a_{13}} & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -\frac{a_{13}\delta}{\beta} & a_{13} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ -\alpha & -\beta & 0 \\ -\gamma + \frac{a_{11}}{a_{13}} & -\delta + \frac{a_{12}}{a_{13}} & 1 \end{pmatrix}.$$

(iii) If $\beta \neq 0$, $\det A \neq 0$ and $v = a_{13}(\det A)/\beta$, then

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \frac{a_{23}}{a_{13}} - \alpha & 1 & 0 \\ \frac{a_{33}}{a_{13}} - \gamma & \frac{\delta}{\beta} & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -\frac{a_{13}}{v} \\ 0 & 1 & -\frac{2a_{13}}{v} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\beta & 0 \\ \frac{v(1-a_{11})}{a_{13}} & -\frac{va_{12}}{a_{13}} & -v \end{pmatrix}.$$

Proposition 2.3. Suppose $a_{13} = 0$. Let further $a_{11} = 0$ and $a_{12} = 0$.

(i) If $a_{33} \neq 0$, then

$$A = \begin{pmatrix} 0 & 0 & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{a_{23}}{a_{33}} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ a_{21} - \frac{a_{23}a_{31}}{a_{33}} & a_{22} - \frac{a_{23}a_{32}}{a_{33}} & 0 \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$$

(ii) If $a_{33} = 0$, then

$$\begin{aligned} A &= \begin{pmatrix} 0 & 0 & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ -a_{31} & -a_{32} & 0 \\ a_{21} + a_{31} & a_{22} + a_{32} & a_{23} \end{pmatrix}. \end{aligned}$$

Proposition 2.4. Suppose $a_{13} = 0$. Let further $a_{12} = 0$ and $a_{11} \neq 0$. Then:

(i) If $a_{23} = 0$, then $A \in \mathcal{L}$.

(ii) If $a_{23} \neq 0$ and $\det A \neq 0$, then putting $a = \det A/a_{11}$ we have

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \frac{a_{21}}{a_{11}} & 1 & 0 \\ \frac{a_{31}}{a_{11}} & \frac{a_{33}-a}{a_{23}} & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{a_{23}}{a} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_{11} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{a(a_{22}-1)}{a_{23}} & a \end{pmatrix}.$$

(iii) If $a_{23} \neq 0$, $\det A = 0$ and $a_{33} \neq 0$, then

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{a_{23}}{a_{33}} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_{11} & 0 & 0 \\ a_{21} + \frac{a_{11}a_{23}}{a_{33}} - \frac{a_{23}a_{31}}{a_{33}} & 0 & 0 \\ a_{31} - a_{11} & a_{32} & a_{33} \end{pmatrix}.$$

(iv) If $a_{23} \neq 0$, $\det A = 0$, and $a_{33} = 0$, these imply $a_{32} = 0$, then

$$A = \begin{pmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & 0 & 0 \end{pmatrix}$$

and

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_{11} & 0 & 0 \\ -a_{31} & 0 & 0 \\ a_{31} + a_{21} & a_{22} & a_{23} \end{pmatrix}.$$

Proposition 2.5. Suppose $a_{13} = 0$. Let further $a_{11} = 0$, $a_{12} \neq 0$. Then:

(i) If $a_{23} = 0$,

$$A = \begin{pmatrix} 0 & a_{12} & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

and

$$A = \begin{pmatrix} a_{12} & 0 & 0 \\ a_{22} - a_{21} & a_{21} & 0 \\ a_{32} - a_{31} & a_{31} & a_{33} \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(ii) Suppose $a_{23} \neq 0$ and $\det A \neq 0$. Putting $\Delta = \det A$ we have

$$A = \begin{pmatrix} 0 & a_{12} & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

and

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \frac{a_{12}(a_{21}+a_{22})-(a_{12}-1)}{a_{12}^2} & 1 & 0 \\ \frac{a_{12}a_{23}(a_{31}+a_{32})-(a_{12}-1)(a_{33}-\Delta)}{a_{12}^2a_{23}} & \frac{a_{33}-\Delta}{a_{23}} & 1 \end{pmatrix} \\ \times \begin{pmatrix} 1 & a_{12} & 0 \\ 0 & 1 & \frac{a_{23}}{\Delta} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -\frac{1}{a_{12}} & 1 & 0 \\ \frac{\Delta(a_{12}a_{21}+1)}{a_{12}a_{23}} & \frac{-\Delta(a_{12}a_{21}+1)}{a_{12}a_{23}} & \Delta \end{pmatrix}.$$

(iii) Suppose $a_{23} \neq 0$, $\det A = 0$ and $a_{21} = 0$. Then $a_{31} = 0$,

$$A = \begin{pmatrix} 0 & a_{12} & 0 \\ 0 & a_{22} & a_{23} \\ 0 & a_{32} & a_{33} \end{pmatrix}$$

and

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \frac{a_{22}-1}{a_{12}} & 1 & 0 \\ \frac{a_{23}a_{32}-(a_{33}-1)}{a_{12}a_{23}} & \frac{a_{33}-1}{a_{23}} & 1 \end{pmatrix} \begin{pmatrix} 1 & a_{12} & 0 \\ 0 & 1 & a_{23} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(iv) Suppose $a_{23} \neq 0$, $\det A = 0$, $a_{21} \neq 0$ and $a_{31} = 0$. Then $a_{33} = 0$,

$$A = \begin{pmatrix} 0 & a_{12} & 0 \\ a_{21} & a_{22} & a_{23} \\ 0 & a_{32} & 0 \end{pmatrix}$$

and

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \frac{a_{22}-1}{a_{12}} & 1 & 0 \\ \frac{a_{23}a_{32}+1}{a_{12}a_{23}} & -\frac{1}{a_{23}} & 1 \end{pmatrix} \begin{pmatrix} 1 & a_{12} & 0 \\ 0 & 1 & a_{23} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(v) If $a_{23} \neq 0$, $\det A = 0$, $a_{21} \neq 0$, and $a_{31} \neq 0$. Then $a_{21}a_{33} = a_{23}a_{31}$ and $a_{33} \neq 0$. Then

$$A = \begin{pmatrix} 0 & a_{12} & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

and

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \frac{a_{22}-1}{a_{12}} & 1 & 0 \\ \frac{a_{32}}{a_{12}} & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a_{12} & 0 \\ 0 & 1 & \frac{a_{23}}{a_{33}} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ a_{31} & 0 & a_{33} \end{pmatrix}.$$

Proposition 2.6. Suppose $a_{13} = 0$. Let further $a_{11} \neq 0$, and $a_{12} \neq 0$. Then:

(i) If $a_{23} = 0$ then

$$A = \begin{pmatrix} a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

and

$$A = \begin{pmatrix} a_{12} & 0 & 0 \\ a_{22} - \frac{a_{21}}{a_{11}} + \frac{a_{22}}{a_{12}} & \frac{a_{21}}{a_{11}} - \frac{a_{22}}{a_{12}} & 0 \\ a_{32} - \frac{a_{31}}{a_{11}} + \frac{a_{32}}{a_{12}} & \frac{a_{31}}{a_{11}} - \frac{a_{32}}{a_{12}} & a_{33} \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -a_{11} & 0 & 0 \\ a_{11} + \frac{a_{11}}{a_{12}} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(ii) If $a_{23} \neq 0$ and $\Delta = \det A \neq 0$ then

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \frac{a_{12}a_{21}-(a_{11}-1)}{a_{11}a_{12}} & 1 & 0 \\ \frac{a_{12}a_{23}a_{31}-(a_{11}-1)(a_{33}-\Delta)}{a_{11}a_{12}a_{23}} & \frac{a_{33}-\Delta}{a_{23}} & 1 \end{pmatrix} \\ \times \begin{pmatrix} 1 & a_{12} & 0 \\ 0 & 1 & \frac{a_{23}}{\Delta} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ \frac{a_{11}-1}{a_{12}} & 1 & 0 \\ 0 & \frac{\Delta(a_{11}a_{22}-a_{12}a_{21}-1)}{a_{11}a_{23}} & \Delta \end{pmatrix}.$$

Proposition 2.7. Suppose $a_{13} = 0$. Let further $a_{11} \neq 0$, $a_{12} \neq 0$, $a_{23} \neq 0$ and $\det A = 0$.

(i) If $a_{11}a_{22} = a_{12}a_{21}$ then $a_{11}a_{32} = a_{12}a_{31}$,

$$A = \begin{pmatrix} a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

and

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \frac{a_{22}-1}{a_{12}} & 1 & 0 \\ \frac{a_{23}a_{32}-(a_{33}-1)}{a_{12}a_{23}} & \frac{a_{33}-1}{a_{23}} & 1 \end{pmatrix} \begin{pmatrix} 1 & a_{12} & 0 \\ 0 & 1 & a_{23} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ \frac{a_{11}}{a_{12}} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(ii) If $a_{11}a_{22} \neq a_{12}a_{21}$ and $a_{11}a_{32} = a_{12}a_{31}$, then $a_{33} = 0$

$$A = \begin{pmatrix} a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & 0 \end{pmatrix}$$

and

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \frac{a_{22}-1}{a_{12}} & 1 & 0 \\ \frac{a_{23}a_{32}+1}{a_{12}a_{23}} & \frac{-1}{a_{23}} & 1 \end{pmatrix} \begin{pmatrix} 1 & a_{12} & 0 \\ 0 & 1 & a_{23} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ \frac{a_{11}}{a_{12}} & 1 & 0 \\ \frac{a_{12}a_{21}-a_{11}a_{22}}{a_{12}a_{23}} & 0 & 1 \end{pmatrix}.$$

(iii) If $a_{11}a_{22} \neq a_{12}a_{21}$, $a_{11}a_{32} \neq a_{12}a_{31}$ then $a_{33} \neq 0$

$$A = \begin{pmatrix} a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

and

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \frac{a_{22}-1}{a_{12}} & 1 & 0 \\ \frac{a_{32}}{a_{12}} & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a_{12} & 0 \\ 0 & 1 & \frac{a_{23}}{a_{33}} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ \frac{a_{11}}{a_{12}} & 1 & 0 \\ \frac{a_{12}a_{31}-a_{11}a_{32}}{a_{12}} & 0 & a_{33} \end{pmatrix}.$$

3. Remarks

Remark 3.1.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \notin \mathcal{LU}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \notin \mathcal{UL}.$$

This is easy to check. Hence we do need three matrices in Theorem 1.3.

Remark 3.2. It is well known that in the case of $\mathcal{LU}$ decompositions of a nonsingular matrix, the choice of the diagonal element of L and U fix the decomposition completely, see [3, Theorem 1, p. 33]. The situation is different in the case of $\mathcal{LU}$ decompositions. In fact we can find a nonsingular matrix A with $A = L_1 U_1 = U_2 M_2$, with all diagonal entries equal to 1.

$$L_1 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad U_1 = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$U_2 = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Remark 3.3. Let

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Then A cannot be expressed as BCD where $B, D \in \mathcal{L}$, $C \in \mathcal{U}$, diagonal entries of B and D are equal to 1.

This can be verified by an explicit computation.

Remark 3.4. However in accordance with Theorem 1.3 we have

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Remark 3.5. Regarding $\mathcal{L}\mathcal{U}$ and $\mathcal{U}\mathcal{L}$ decompositions we state the following results. Let $A \in M_n(F)$ be nonsingular, $n \geq 2$. Then the following conditions are equivalent.

- (1) There exists $P \in \mathcal{U}$ with $p_{ii} = 1$ for all i such that $B = AP \in \mathcal{L}$.
- (2) $A = LU$ for some $L \in \mathcal{L}$, $U \in \mathcal{U}$.
- (3) $\Delta_s(A) \neq 0$ for all s with $1 \leq s \leq n$, where $\Delta_s(A) = \det(a_{ij})$ ($1 \leq i, j \leq s$).

The equivalence of (1) and (3) is proved in [4], the equivalence of (1) and (2) can be proved using the diagonal trick of Theorem 1.3 and the fact that if $U \in \mathcal{U}$ with unit diagonal then $U^{-1} \in \mathcal{U}$ with unit diagonal.

Following the referee's suggestion, we introduce the matrix

$$T = \begin{pmatrix} 0 & 0 & \cdots & \cdots & 1 \\ 0 & 0 & \cdots & 1 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Then $T^2 = I$, so the function $A \rightarrow TAT$ is an automorphism on $M_n(F)$. Also

$$TAT = \begin{pmatrix} a_{nn} & \cdots & \cdots & \cdots & a_{n1} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{2n} & \cdots & \cdots & a_{22} & a_{21} \\ a_{1n} & \cdots & \cdots & a_{12} & a_{11} \end{pmatrix}.$$

Hence if $A \in \mathcal{U}$ (or $\mathcal{L}$) then $TAT \in \mathcal{L}$ (or $\mathcal{U}$). Let $\delta_s(A) = \det(a_{ij})$, $s \leq i \leq n$, $s \leq j \leq n$. Then $\delta_s(A) = \Delta_s(TAT)$ for each integer s . Hence this automorphism converts LU factorization into UL factorization and converts 'upper left' principal minors into 'lower right' minors. Hence we can deduce the following from the result mentioned at the beginning of this remark.

Theorem 3.6. Let $A \in M_n(F)$ be nonsingular, $n \geq 2$. The following conditions are equivalent.

- (a) There exists a $P \in \mathcal{U}$ with $p_{ii} = 1$ for all i such that $B = PA \in \mathcal{L}$.
- (b) $A = UL$ for some $U \in \mathcal{U}$, $L \in \mathcal{L}$.
- (c) $\delta_s(A) \neq 0$, $1 \leq s \leq n$.

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